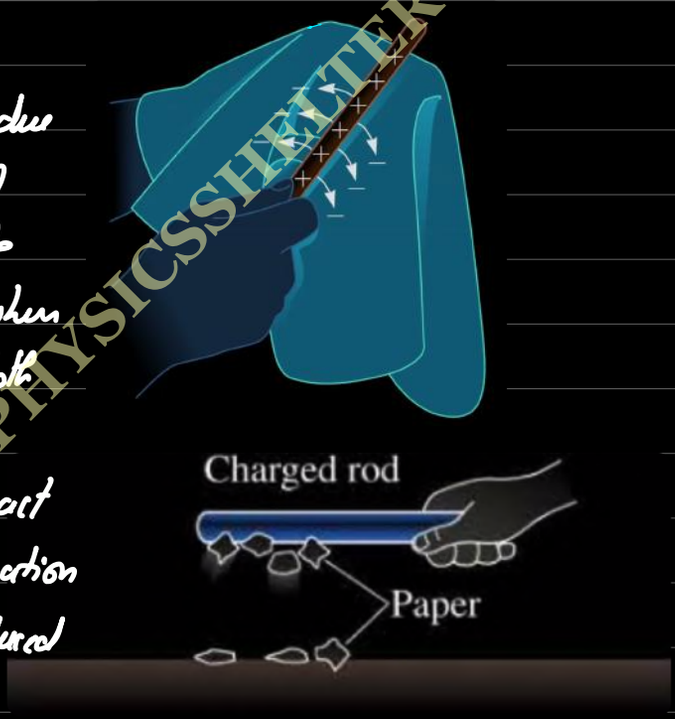
ELECTRIC CHARGE: Charge is a fundamental property of electrons and protons by virtue of which material objects participate in electric interactions (Apply forces) on each other.

PROPERTIES OF CHARGE:

- Charge is quantized (Always in multiples of 1.6×10^{-19} coulombs).
- Always conserved including nuclear interactions. [Even though positive charge can destroy negative charge in pair annihilation but net charge still remains same algebraically].
- Like charges repel unlike charges attract.

CHARGING BODIES BY FRICTION: When substances are rubbed together then due to differing electron affinities, one of the substance may lose electrons while another may gain. For instance when a glass rod is rubbed with a silk cloth it loses electrons to the cloth and gets positively charged and can attract paper chits due to induction (separation of positive and negative charges induced by presence of charges in surroundings).

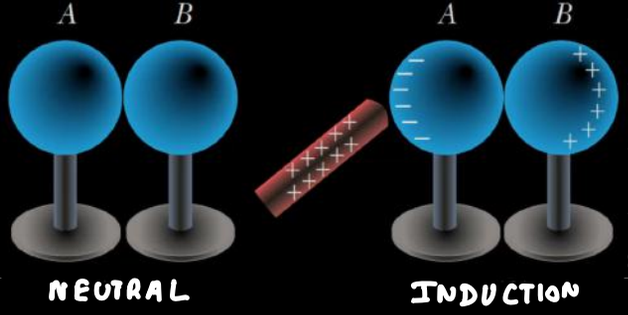


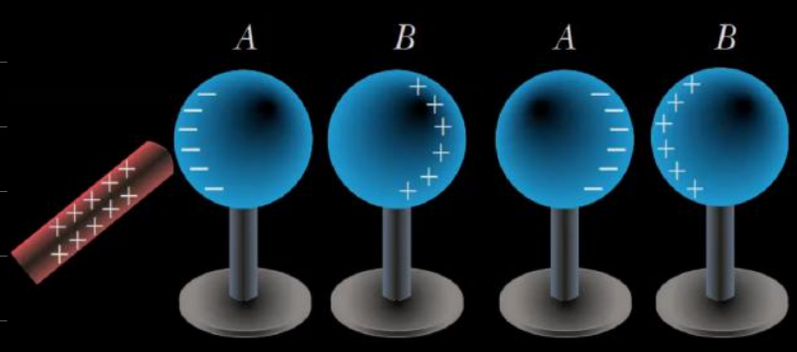
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CONDUCTORS: In metals, certain electrons are not bonded to a single nucleus, rather they are free to move in the entire metal matrix freely (such property might be found in some non metals like graphite as well). This type of substances are called conductors, materials without this property are called insulators.

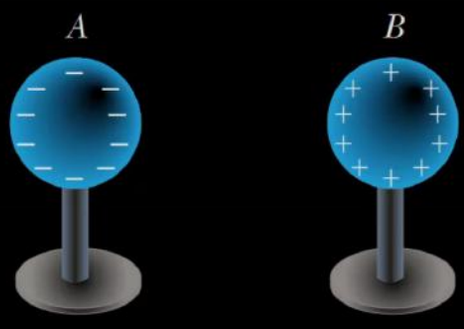
CHARGING CONDUCTING PAIR BY INDUCTION:

Due to presence of free electrons, conductors can be easily charged as illustrated using the process of induction.





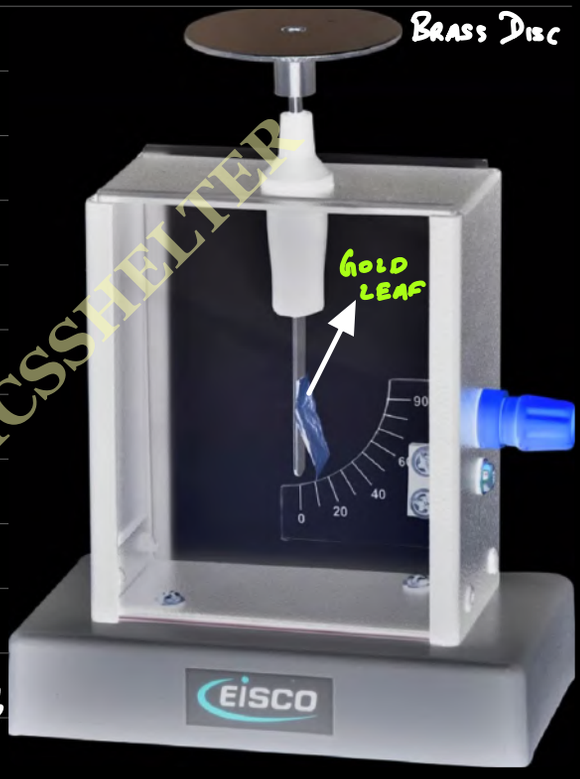
SEPARATION



ROD REMOVED

REMOTE CHARGED SPHERES.

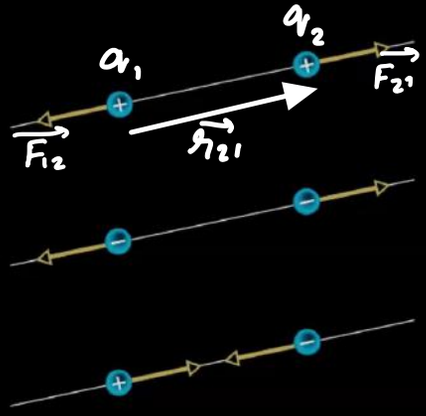
GOLD LEAF ELECTROSCOPE: It is a device for measuring the charges of charged bodies. There is a brass conductor at the top which is connected to two gold leaves. When a charged body is brought in contact with the brass conductor some charge flows from it in to gold leaves thus creating a repulsion between them. Due to this gold leaf deflects which can be calibrated against etchings on a circular arc to find relative magnitude of charge transferred.



(Gold leaf electroscope was used to infer photoelectric effect due to irradiation of metals with ultraviolet light).

COULOMB'S LAW: Every charged particle in the universe applies a force on every other charged particle and this force is given by:

$$\vec{F}_{21} = \frac{k\, q_1 q_2}{r_{21}^2}\,\hat{r}_{21}$$



Alternatively:

$$\vec{F}_{21} = \frac{k\, q_1 q_2\, \vec{r}_{21}}{r_{21}^3}$$

← Force on 2 due to 1

$k = 9\times10^9\ N m^2/C^2 = \frac{1}{4\pi\epsilon_0}$

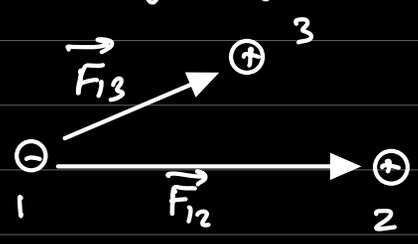
$\epsilon_0 \rightarrow$ Permittivity of free space.

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SUPERPOSITION PRINCIPLE: Force of interaction between any two charged particles is independent of presence or absence of any other charge in the universe.

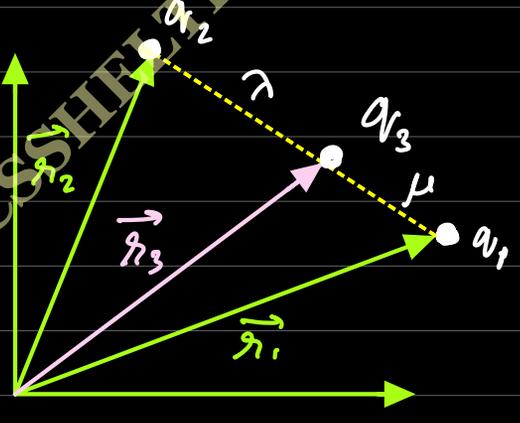
Thus $\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13}$



EXAMPLE (IRODOV 3·4) Two positive charges q_1 and q_2 are located at position vectors $\vec{r}_1$ and $\vec{r}_2$ respectively. Find a negative charge q_3 and its position vector $\vec{r}_3$ such that net force acting on each charge is zero.

Solution: Clearly the q_3 must lie on line joining q_1 and q_2 for vector sum of two forces on it to be zero.

Let $\vec{r}_3$ divide line joining q_2 and q_1 in ratio $\lambda : \mu$. Then for force balance, $\frac{q_2}{\lambda^2} = \frac{q_1}{\mu^2} \Rightarrow \frac{\lambda}{\mu} = \sqrt{\frac{q_2}{q_1}}$



Thus $\vec{r}_3$ divides line joining q_2 and q_3 in ratio $\sqrt{q_2} : \sqrt{q_1}$

Now by section formula:

$$\vec{r}_3 = \frac{\lambda \vec{r}_1 + \mu \vec{r}_2}{\lambda + \mu}$$

$$\Rightarrow \vec{r}_3 = \frac{\sqrt{q_2}\,\vec{r}_1 + \sqrt{q_1}\,\vec{r}_2}{\sqrt{q_2} + \sqrt{q_1}} \quad \text{Ans.}$$

Now q_3 must be negative so that it can balance the repulsion between positive charges. Thus:

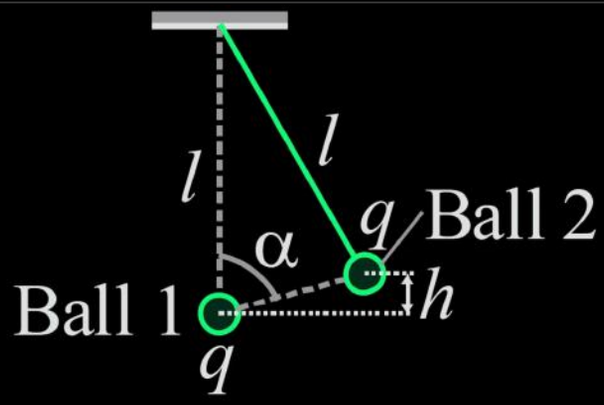
$$\frac{-q_3 q_2}{\lambda^2} = \frac{q_1 q_2}{(\lambda+\mu)^2}$$

$$\Rightarrow q_3 = -q_1 \left(\frac{\lambda}{\lambda+\mu}\right)^2 = -q_1 \left(\frac{\sqrt{q_2}}{\sqrt{q_2}+\sqrt{q_1}}\right)^2$$

$$\Rightarrow q_3 = \frac{-q_1 q_2}{(\sqrt{q_1}+\sqrt{q_2})^2} \quad \text{Ans.}$$

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EXAMPLE: In the shown figure, ball 1 is fixed while ball 2 has mass m and is in equilibrium. Find the angle α in terms of mg, q and l. Also find the value of h in these terms.

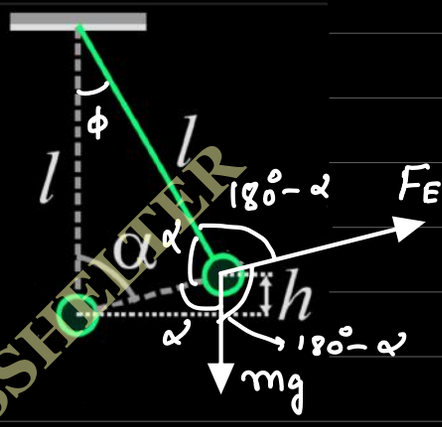


Solution: By Lami's theorem:

$$\frac{mg}{\sin\alpha} = \frac{F_E}{\sin 2\alpha} = \frac{T}{\sin\alpha} = \frac{kq^2}{(2l\cos\alpha)^2 \sin 2\alpha}$$

$$\Rightarrow \frac{mg\,(4l^2\cos^2\alpha)\sin 2\alpha}{\sin\alpha} = kq^2$$

$$\Rightarrow 8mgl^2\cos^3\alpha = kq^2 \Rightarrow \cos\alpha = \left(\frac{kq^2}{8mgl^2}\right)^{1/3}$$



Now $h = l[1-\cos\phi]$ also $\phi = (180^\circ - 2\alpha)$

$$\Rightarrow h = l[1+\cos 2\alpha] = 2l\cos^2\alpha = 2l\left(\frac{kq^2}{8mgl^2}\right)^{2/3}$$

POINT CHARGES SEPARATED BY MATERIAL MEDIA: When point charges are placed in material media, the material medium gets polarized (positive and negative charge centres separated) due to interaction with the immersed charges, thus the point charges not only experience force due to each other but also force due to polarization of the material medium and net force on each one reduces by a factor K known as dielectric constant of the medium about which we shall study in great detail when we study dielectrics. Thus if F_0 were the force experienced by each charge due to electrostatic interaction with each other then net electrostatic force in presence of material medic is given as:

$$\vec{F}_{net} = \frac{\vec{F}_0}{K}$$ where K is dielectric constant of material medium.

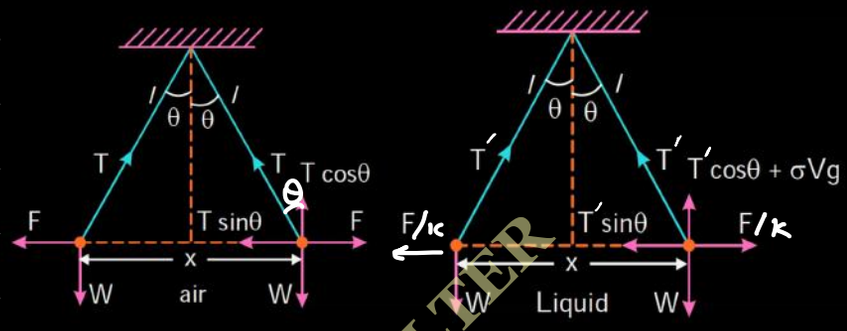
→ Note that this does not violate superposition principle. The force due to each other on the charges is still $\vec{F}_0$, however now they are also experiencing force due to polarized material media.

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→ From above statements it also follows that a separation of r_0 in a material medium of dielectric constant k is equivalent to a separation of $r_0\sqrt{k}$ between two charges for purpose of calculating net electrostatic forces on charges.

EXAMPLE:

Two identical balls each having density ρ are suspended from a common point by two insulating strings of equal length Both the balls have equal mass and charge. In equilibrium, each string makes an angle θ with the vertical. Now both the ball are immersed in a liquid. As a result, the angle θ does not change. The density of the liquid is σ. Find the dielectric constant of the liquid.



Solution: Refer to the figures above: By Lami's theorem, (in air):

$$\frac{F}{\sin(180°-\theta)} = \frac{W}{\sin(90°+\theta)} \quad (1)$$

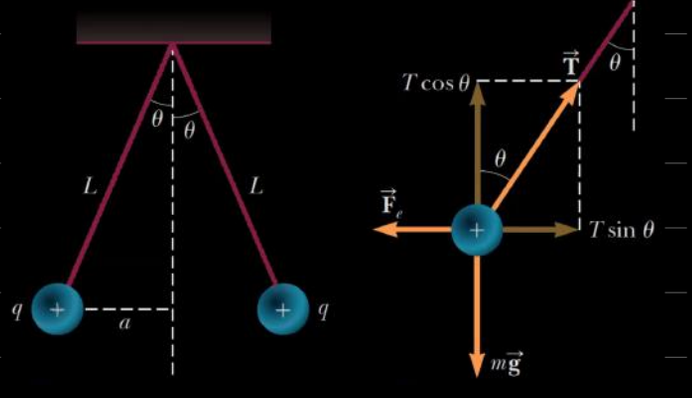
Upon immersing in liquid:

$$\frac{F/k}{\sin(180°-\theta)} = \frac{W-\sigma Vg}{\sin(90°+\theta)} \quad (2)$$

Now (1)/(2)

$$\Rightarrow k = \frac{W}{W-\sigma Vg} = \frac{\rho Vg}{\rho Vg-\sigma Vg} = \rho/(\rho-\sigma) \text{ Ans.}$$

EXAMPLE: Two point charges are hanging from identical threads in equilibrium as shown find the charge q in terms of angle θ, l and mass m of the charges.



Solution: By Lami's theorem:

$$\frac{mg}{\sin(90°+\theta)} = \frac{kq^2}{(2l\sin\theta)^2\sin(180°-\theta)}$$

$$\Rightarrow q = \sqrt{mg\tan\theta\,(2l\sin\theta)^2/k} \text{ Ans.}$$

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ELECTRIC FIELD (INTENSITY) : Electrostatic force experienced per unit charge kept at any point of space is called electric field intensity at that point. Mathematically if $d\vec{F}$ is the force experienced by a charge dq placed at some point of space then, the electric field at that point is given as:

$$\vec{E} = d\vec{F}/dq \quad \text{or} \quad \boxed{\vec{E} = \vec{F}/q}$$

this is more often used in the form:

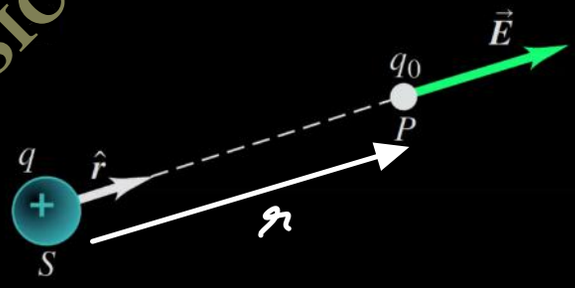
$$\boxed{\vec{F} = q\vec{E}}$$

Thus to find electric field at any point we can imagine a unit charge kept at that point and simply find electrostatic force on it.

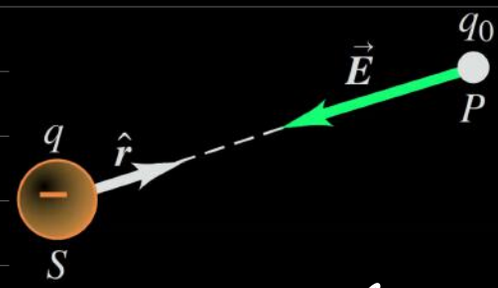
ELECTRIC FIELD DUE TO A POINT CHARGE :

$$\vec{F} = \frac{k\,q q_0}{r^2}\hat{r} \Rightarrow \vec{E} = \frac{\vec{F}}{q_0} = \frac{kq}{r^2}\hat{r}$$

$$\Rightarrow \boxed{\vec{E} = \frac{kq}{r^2}\hat{r}}$$



For positive charges field is directed away from the charge while for negative charges, field is directed towards the charge.



ELECTRIC FIELD LINES : Electric field lines are imaginary lines drawn to visualize electric field around a charge configuration. Following are some properties of electrostatic electric field lines:

1. Their density is proportional to strength of electric field and their tangent gives the direction of electric field.
2. They emanate from positive charges or infinity and terminate at a negative charge or infinity.
3. Number of field lines emanating from a charge is proportional to the value of the charge.
4. Electric field lines never intersect each other (∵ otherwise

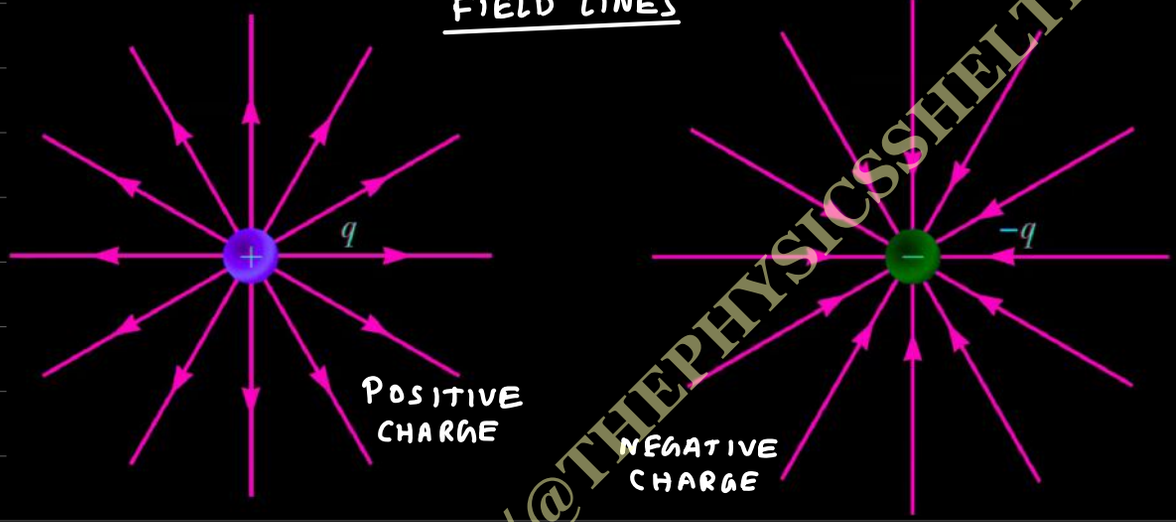
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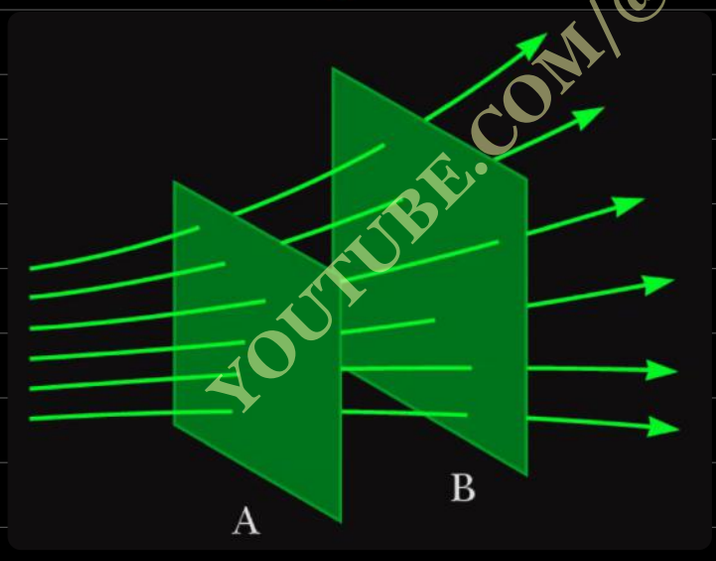
tangent at intersection point will not be uniquely defined however any material particle must experience force in a unique direction.

(5) Electrostatic Electric field lines never form closed loops (∵ Electrostatic field is conservative ∴ work done by it in a closed loop must be zero).

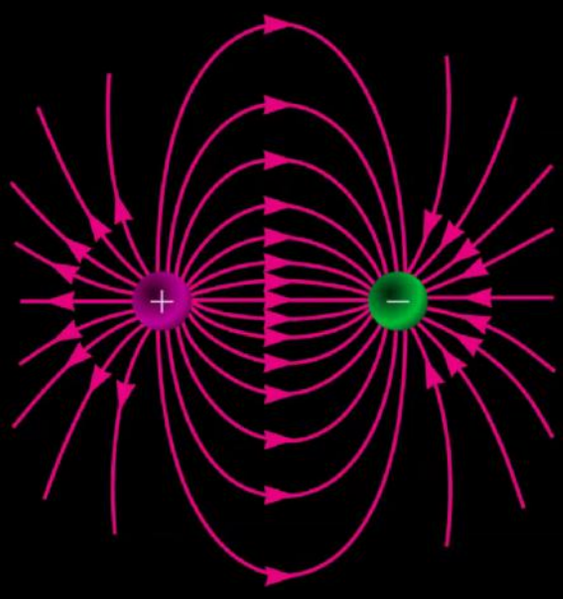
(6) At conducting surfaces, Electric field lines always terminate or emanate along normal to the surface.

FIELD LINES



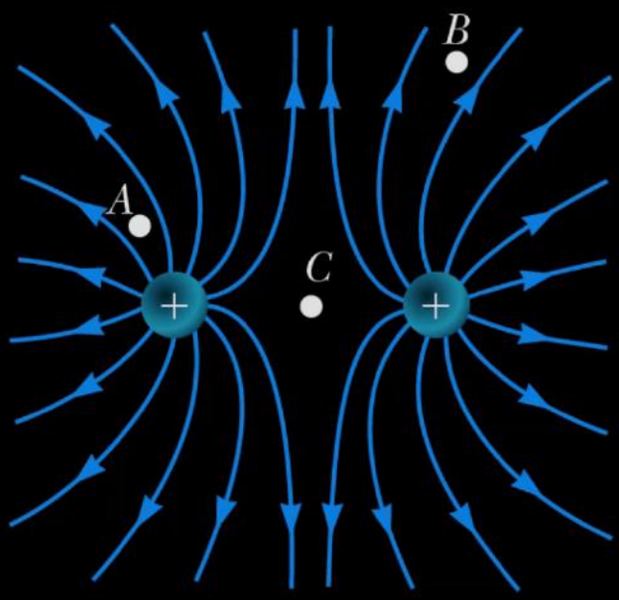


FIELD AT A IS GREATER THAN FIELD AT B.

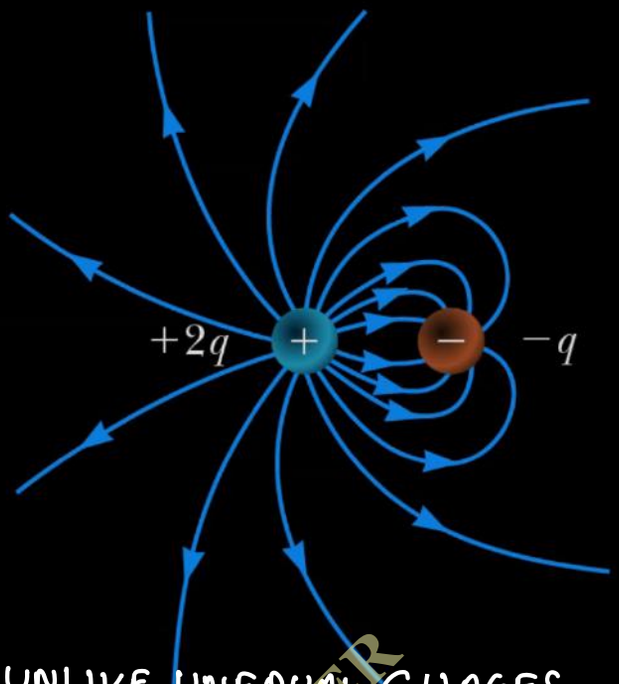


FIELD DUE TO DIPOLE (UNLIKE EQUAL MAGNITUDE CHARGES)

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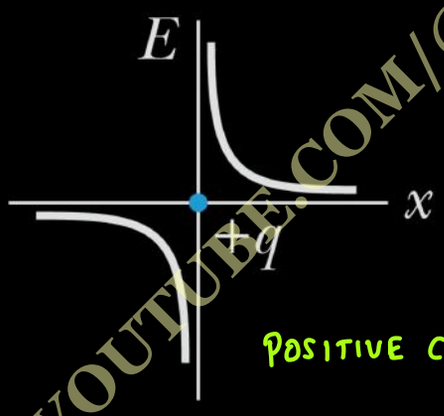
LIKE EQUAL CHARGES



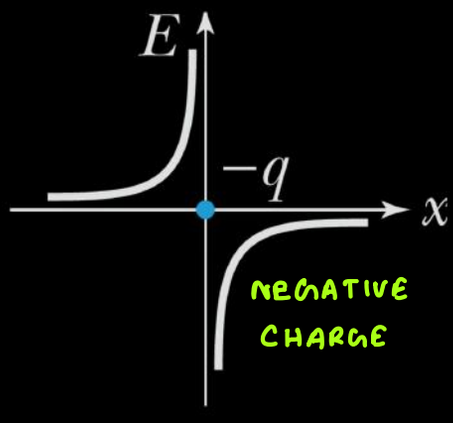
UNLIKE UNEQUAL CHAGES

→ Notice the ratio of number of field lines associated with various charges in above figures. (This is also the ratio of charges).

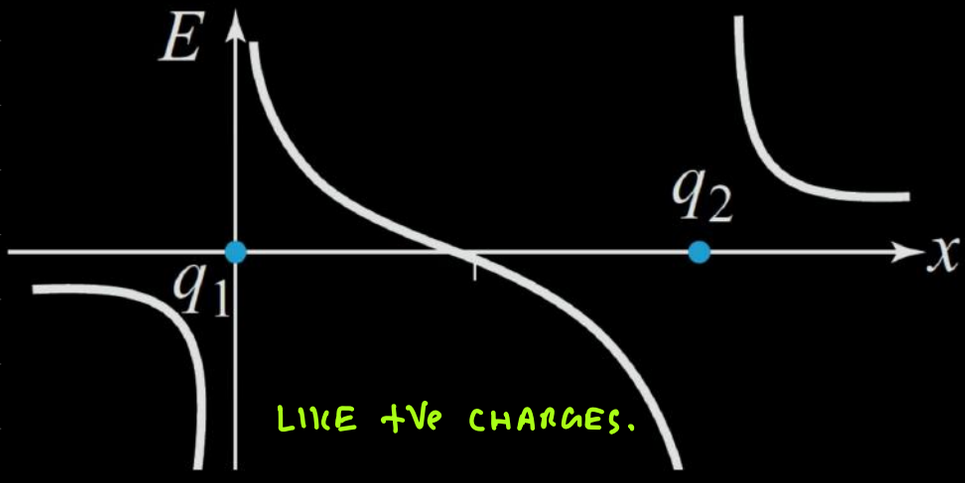
FIELD Vs x PLOTS FOR SOME POINT CHARGE GEOMETRIES:



POSITIVE CHARGE

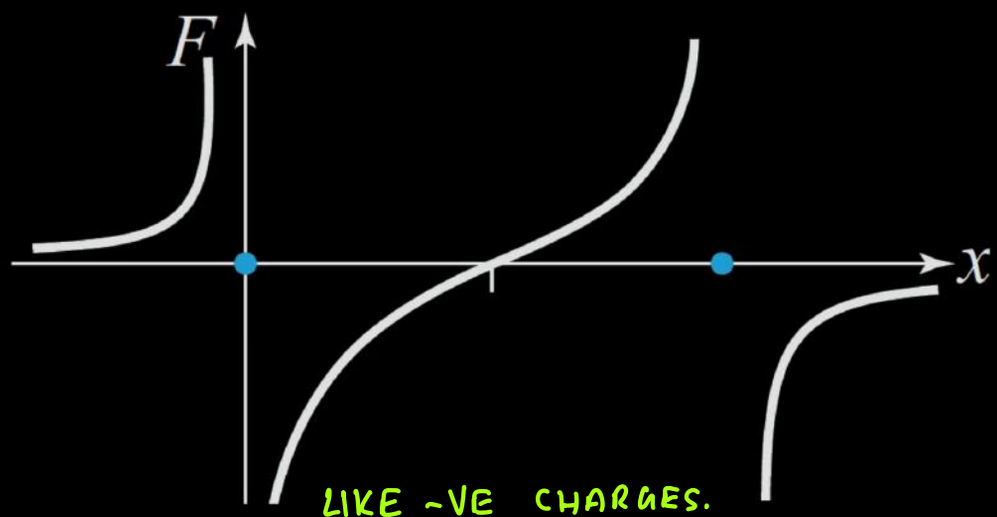


NEGATIVE CHARGE



LIKE +Ve CHARGES.

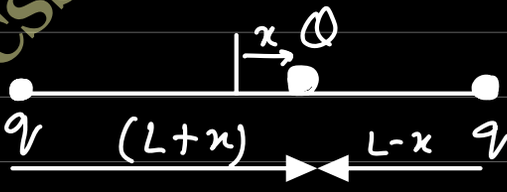
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LIKE -VE CHARGES.

EXAMPLE: Point charge Q of mass m is placed at mid point of two like charges q each separated by a distance $2L$. Find time period of its small oscillations. (Charges q are fixed).

Solution: Let Q be displaced by a small amount x from mid point. Now



Restoring force:

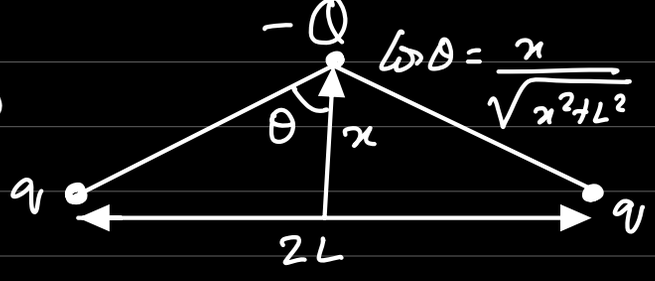
$$F = kQq\left[\frac{1}{(L-x)^2} - \frac{1}{(L+x)^2}\right] = kQq\,\frac{4Lx}{(L^2-x^2)^2}$$

$x << L \Rightarrow F \approx 4kQqx/L^3 = -m\,d^2x/dt^2$ (SHM!)

$$\Rightarrow T = 2\pi\sqrt{\frac{mL^3}{4kQq}} \quad \text{Ans.}$$

EXAMPLE: Repeat the previous problem if the charge were instead $-Q$ and displacement were given perpendicular to the line joining charges of value q.

Solution: Restoring Force: $\dfrac{2kQq}{(L^2+x^2)}\cdot\cos\theta$



$$\Rightarrow F(x) = \frac{2kQq}{(L^2+x^2)}\cdot\frac{x}{\sqrt{x^2+L^2}}$$

But $x << L \Rightarrow F = \dfrac{2kQqx}{L^3} = -m\dfrac{d^2x}{dt^2} \Rightarrow T = 2\pi\sqrt{mL^3/2kQq}$ Ans.

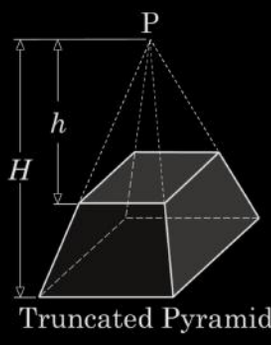
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Example (Pathfinder Build 25)

A right pyramid of square base and height H has uniform charge distributed everywhere. Magnitude of electric field due to this at apex is E_0. Now a symmetrical portion of height h is removed. Find the electric field due to the remaining truncated pyramid.



Pyramid

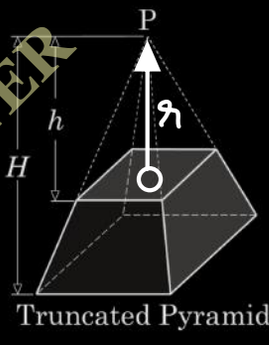


Truncated Pyramid

Solution: Consider similar volumes elements in the original pyramid and removed pyramid. The ratio of volumes must be proportional to cube of the linear dimensions, thus $q_h / q_H = h^3/H^3$, Also the corresponding distances from apex must be in the ratio of linear dimensions. $\therefore R/r = H/h$. $\therefore$ Ratio of their Electric field contributions:



Pyramid



Truncated Pyramid

$$\frac{E_h}{E_H} = \frac{k\, q_h / r^2}{k\, q_H / R^2} = \frac{h^3}{H^3} \times \frac{H^2}{h^2} = \frac{h}{H}$$

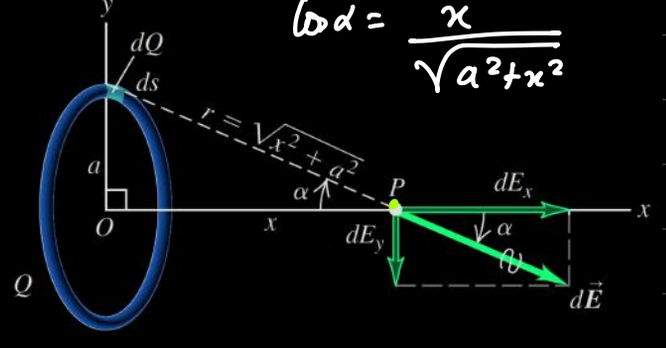
Since this ratio is constant for every similar pair of Volumes
$\therefore$ this must also be the ratio of net fields.

$$\Rightarrow \frac{E_{rp}}{E_0} = \frac{h}{H} \quad (rp = \text{removed pyramid}).$$

$$\therefore E_{TP} = E_0 - E_{rp} = E_0\left(1 - \frac{h}{H}\right) \text{ Ans.}$$

Field On The Axis Of A Ring:

Consider an element having a charge dQ as shown in the figure. Axial field due to this element is:



$$\cos\alpha = \frac{x}{\sqrt{a^2+x^2}}$$

$$dE_x = \frac{k\, dQ}{(x^2+a^2)} \cdot \cos\alpha$$

$$\Rightarrow dE_x = \frac{k\, dQ\, x}{(x^2+a^2)^{3/2}}$$

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Now integrating this over the entire ring:

$$E_x = \frac{kQx}{(a^2+x^2)^{3/2}}$$

This is also the net field since radial components of field cancel out due to symmetry.

Question: Plot the field due to a uniformly charged ring wrt x, where is the field maximum.

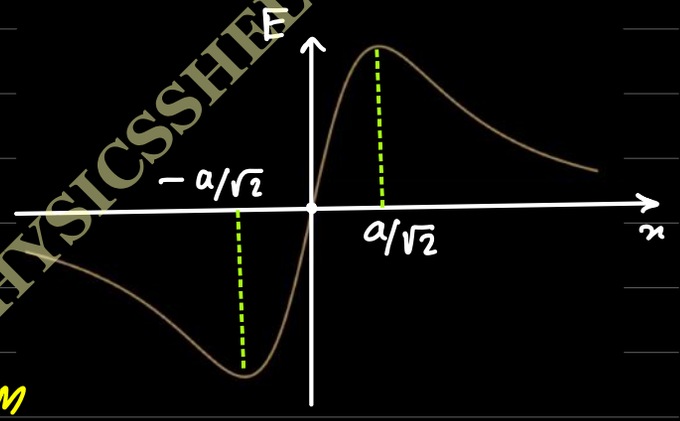
Sol: For maximum field, $\frac{dE_x}{dx} = 0 \Rightarrow \frac{d}{dx}\left[x^{-1}(a^2+x^2)^{3/2}\right] = 0$

$$\Rightarrow x^{-1} \times \frac{3}{2}(a^2+x^2)^{1/2} \cdot 2x + (a^2+x^2)^{3/2} \times (-1)x^{-2} = 0$$

Simplifying, rearranging and solving we get, for maximum field:

$$x = \pm a/\sqrt{2}$$

The plot is shown in adjacent figure,



→ For $x \ll a$, the graph is nearly a straight line, $E_x \approx \frac{kQx}{a^3}$

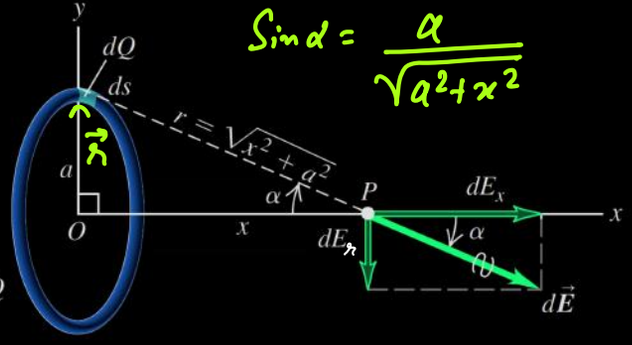
This can be used for working out SHM of charged particles near centre of a charged ring.

→ At $\pm\infty$ field goes to zero.

→ Even if the charge were non uniformly distributed on the ring, the axial component of field would still be the same.

RADIAL FIELD DUE TO NON UNIFORMLY CHARGED RING (ON AXIS):

Consider a charge dQ at position vector $\vec{r}$ as shown, radial field due to this is given as:



$$\sin\alpha = \frac{a}{\sqrt{a^2+x^2}}$$

$$d\vec{E}_r = \frac{k\,dQ}{x^2+a^2} \cdot \frac{a}{\sqrt{x^2+a^2}}(-\hat{r})$$

$$\Rightarrow d\vec{E}_r = \frac{-k}{(x^2+a^2)^{3/2}} \cdot (\vec{r}\,dQ) \Rightarrow \vec{E}_r = \frac{-k}{(x^2+a^2)^{3/2}} \int \vec{r}\,dQ$$

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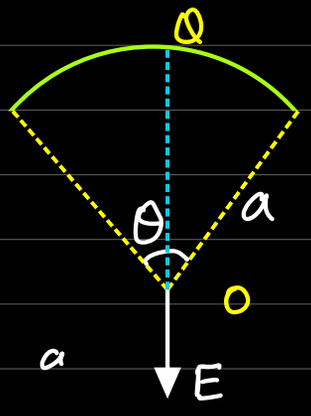
$\Rightarrow \vec{E_n} = \frac{-kQ}{(x^2+a^2)^{3/2}} \cdot \frac{1}{Q}\int \vec{x}\, dQ$, Now $\frac{1}{Q}\int \vec{x}\, dQ$ is centre of charge of the ring (Similar to CM), thus:

$$\boxed{\vec{E_n} = \frac{-kQ}{(x^2+a^2)^{3/2}} \vec{r}_{CQ}}$$

Think what will be the radial field on axis of a semicircular ring carrying a uniformly distributed charge Q.

Ans: $\frac{-kQ}{(x^2+a^2)^{3/2}} \cdot \frac{2a}{\pi}$ $(\because r_{CQ} = 2a/\pi)$.

Electric Field At Centre Of Curvature Of Uniformly Charged Arc:



Here we can find the field at O either using integration of elemental fields (Note that by symmetry, the net field must be along bisector of the arc). Alternatively we can simply put $x=0$ in the result for radial field due to a general ring. (Formula in previous section).

Thus:

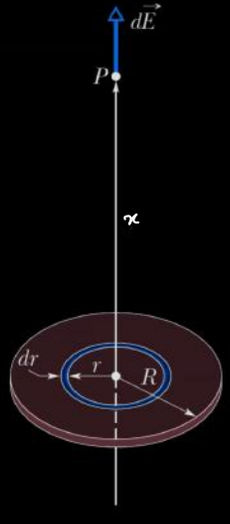
$$E = \frac{kQ}{a^3} \cdot r_{CQ}, \quad \text{Recall CM for arc: } r_{CM} = \frac{R\sin(\theta/2)}{(\theta/2)}$$

$$\Rightarrow E = \frac{kQ}{a^3} \cdot \frac{a\sin(\theta/2)}{(\theta/2)} = \frac{kQ}{a^2}\frac{\sin(\theta/2)}{(\theta/2)}$$

Field On The Axis Of A Disc

: Field due to a disc can be estimated by breaking it in to elemental ring elements and then using the result for ring derived in the previous sections for integrating over the entire disc. Consider a disc with uniform surface charge density. The element at a distance r and of thickness dr has a charge:

$dQ = \sigma \times 2\pi r\, dr$.



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Thus axial field due to this element is given as:

$$dE_x = \frac{k \times \sigma \times 2\pi r\,dr}{(r^2+x^2)^{3/2}} \times x \qquad \left(\because \frac{kQx}{(R^2+x^2)^{3/2}}\right)$$

Thus $E_x = 2\pi k\sigma x \int_0^R \frac{r\,dr}{(r^2+x^2)^{3/2}} = \frac{\sigma}{2\epsilon_0}\left[1 - \frac{x}{\sqrt{x^2+R^2}}\right]$

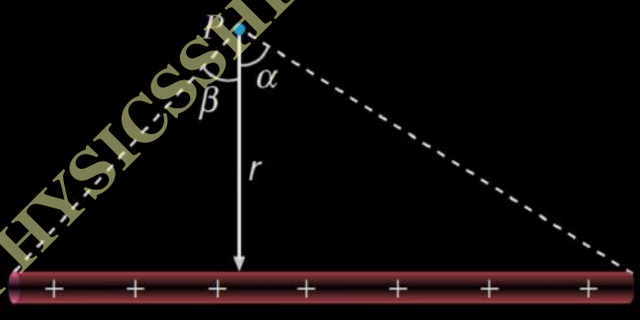
where $k = 1/4\pi\epsilon_0$.

→ By letting $R \to \infty$, we can get field for an infinite sheet.

We get $\boxed{E = \frac{\sigma}{2\epsilon_0}}$ ← FIELD DUE TO INFINITE SHEET.

ELECTRIC FIELD DUE TO A UNIFORMLY CHARGED ROD:

Consider a rod carrying uniform linear charge density λ. We wish to find components of electric field due to this rod parallel and perpendicular to the rod at a point which subtends angles α and β at its extremities.

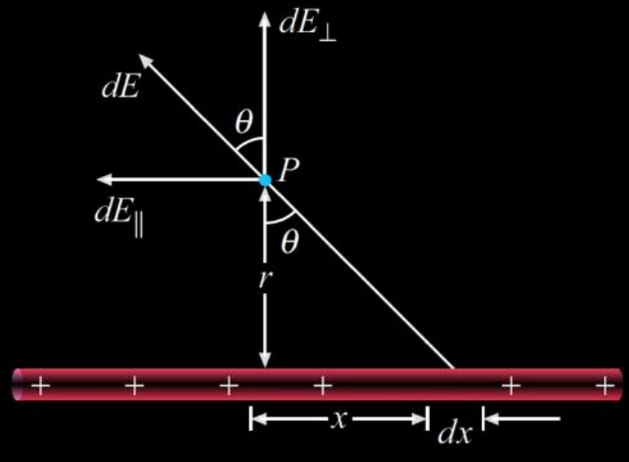


Consider an element of length dx as shown, due to this element, $dE = \frac{k\lambda\,dx}{(r\sec\theta)^2}$. Also $x = r\tan\theta$

$\Rightarrow dx = r\sec^2\theta\,d\theta$.

Now $dE_{\parallel} = dE\sin\theta$ & $dE_{\perp} = dE\cos\theta$

Substituting dx and integrating for $E_{\parallel}$.



$$E_{\parallel} = \int \frac{k\lambda \times r\sec^2\theta\,d\theta}{r^2\sec^2\theta}\sin\theta = \frac{k\lambda}{r}\int_{-\beta}^{\alpha}\sin\theta\,d\theta \qquad (1)$$

Thus $E_{\parallel} = \frac{k\lambda}{r}[\cos\beta - \cos\alpha]$ $(-\hat{i})$

$\Rightarrow$ $\boxed{E_{\parallel} = \frac{k\lambda}{r}[\cos\alpha - \cos\beta]\,\hat{i}}$

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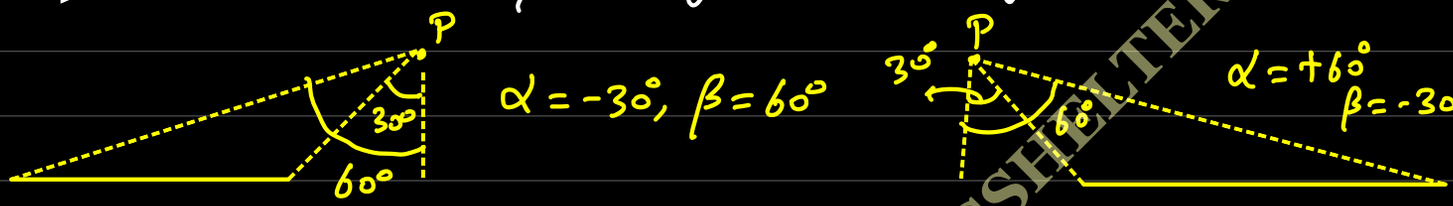
→ Note that here α is the angle subtended in the $\hat{i}$ direction and β is angle subtended in direction $(-\hat{i})$.

Now for $E_\perp$ we just need to replace $\sin\theta$ by $\cos\theta$ in eqn ①

Thus, $E_\perp = \frac{k\lambda}{R}\int_{-\beta}^{\alpha} \cos\theta\, d\theta = \frac{k\lambda}{R}[\sin\alpha + \sin\beta]$

$$\boxed{E_\perp = \frac{k\lambda}{R}[\sin\alpha + \sin\beta]}$$

→ Note that α and β are signed values. e.g.:



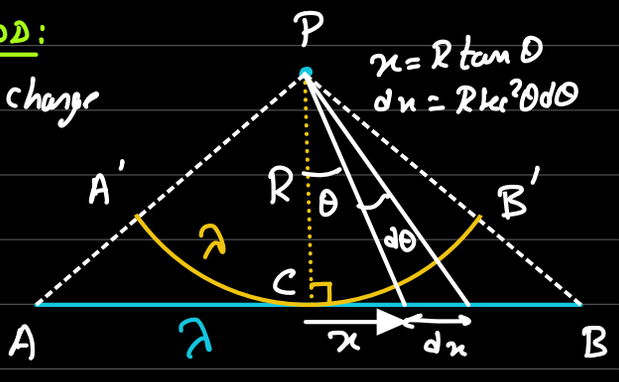
→ If loading from P rod is infinite in both directions then $\alpha = \beta = 90°$

$\Rightarrow E_\parallel = 0$, $\boxed{E_\perp = 2k\lambda/R}$

→ If P is vertically above one of the ends of a semi-infinite rod then $\alpha = 90°$, $\beta = 0°$ $\Rightarrow$ $\boxed{E_\perp = E_\parallel = k\lambda/R}$ In this case net field is inclined at 45° to the rod.

ARC EQUIVALENT OF A UNIFORMLY CHARGED ROD:

If AB is a charged rod having uniform charge density λ we can show that its field at point P will be same as that generated by an arc A'B' having uniform charge density λ. The arc is centred at point P and clipped at radial lines PA and PB. We shall that field due to dx of line segment is same as field due to $d\theta$ of the arc.



Due to dx:

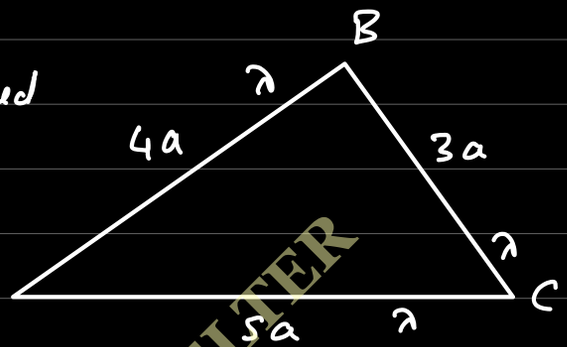
$$dE = \frac{k\lambda\, dx}{(R\sec\theta)^2} = \frac{k\lambda}{R^2\sec^2\theta}\cdot R\sec^2\theta\, d\theta = \frac{k\lambda\, d\theta}{R} \quad ①$$

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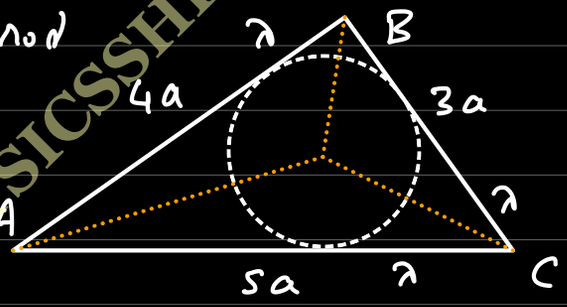
Now due to arc element $R d\theta$, $dE' = \frac{k \times \lambda R d\theta}{R^2} = \frac{k\lambda}{R} d\theta$ (2)

From (1) & (2), $dE = dE'$. Thus corresponding elements of arc and line segment generate identical fields at P ∴ net field due to line segment must be the same as net field due to corresponding equivalent arc. C.: Along bisector of Equivalent Arc).

EXAMPLE: A right triangular frame is formed using three rods each having uniform linear charge density λ. Find the electric field at incircle of the frame.



Solution. Consider the adjacent figure. Each rod can be treated as arc of incircle thus three rods complete the incircle ∴ by symmetry field at incentre is zero.



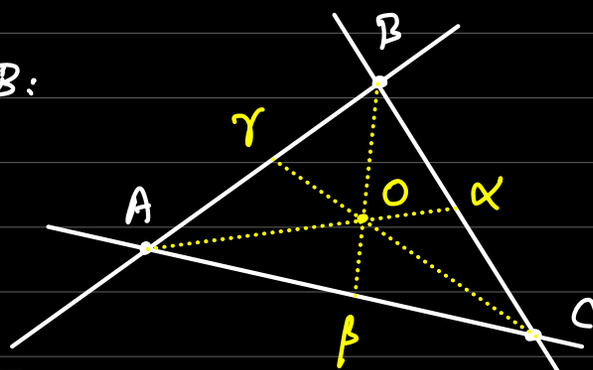
EXAMPLE: If we have three infinite lines each carrying a charge density λ in a plane show that Electric field at the centroid of the triangle formed by their pairwise intersections is zero.

Solution: Let outward unit vector be $\hat{n}$.

Magnitude of field due to line along AB:

$$E_r = 2k\lambda / (h_r/3)$$

Where h_r is altitude on AB of ΔABC.



$$\Rightarrow E_r = \frac{6k\lambda}{h_r} \times \frac{\overline{AB}}{\overline{AB}} = \frac{3k\lambda}{S} \times (\overline{AB}) \quad (\because S = \text{Area of } \Delta = \tfrac{1}{2} AB \times h_r)$$

Vectorially:

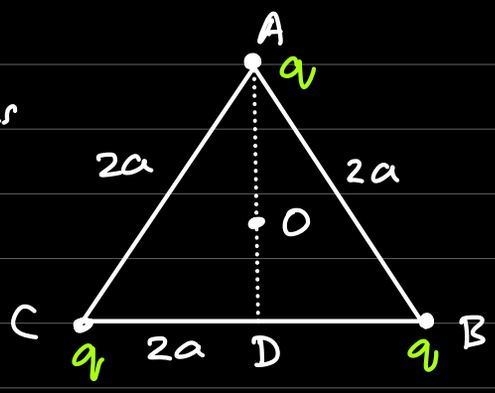
$$\vec{E}_r = \frac{3k\lambda}{S}(\vec{AB} \times \hat{n}), \text{ Similarly } \vec{E}_\alpha = \frac{3k\lambda}{S}(\vec{BC} \times \hat{n})$$

and $\vec{E}_\beta = \frac{3k\lambda}{S}(\vec{CA} \times \hat{n})$, Net field: $\vec{E}_\alpha + \vec{E}_\beta + \vec{E}_r$

$$= \frac{3k\lambda}{S}(\vec{AB} + \vec{BC} + \vec{CA}) \times \hat{n} = 0 \quad \text{QED.}$$

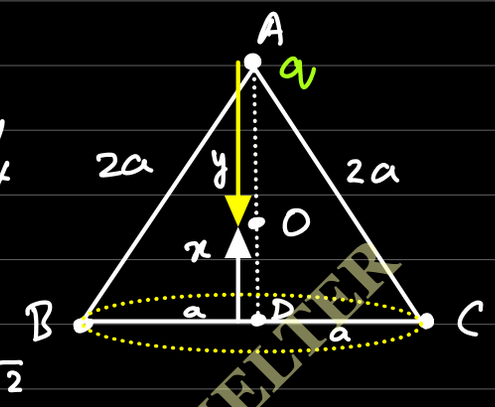
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EXAMPLE: Identical charges q each are placed at vertices of equilateral triangle as shown, show that on the median AD apart from the centroid O, electric field must be zero at some point between O and D also.



Solution: Recall that even for a non uniformly charged ring, axial field is same as that for a uniformly charged ring, which is maximum at $a/\sqrt{2}$.

Now $OD = \frac{1}{3} \times 2a \cos 30° = (a/\sqrt{3}) < a/\sqrt{2}$



Now field on axis of ring: $E = kq\left[x(a^2+x^2)^{-3/2}\right]$, at $x = a/\sqrt{3}$,

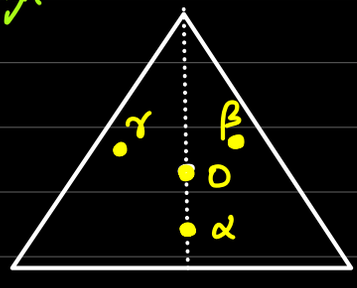
$\frac{dE}{dx} = \frac{3\sqrt{3}}{32}\left(\frac{kq}{a^3}\right)$, For point charge at A, field at $y = kq/y^2$,

$\Rightarrow \left(\frac{dE}{dy}\right)$ mean $y = \frac{2a}{\sqrt{3}}$: $\left|\frac{dE}{dy}\right| = \frac{3\sqrt{3}kq}{4a^3}$, Thus $\left|\frac{dE}{dx}\right| < \left|\frac{dE}{dy}\right|$

Thus as we move from O towards D, field due to ring reduces at a slower rate than that of point charge at A. ∴ Field slightly below O must be towards A, while field at D is clearly downwards ∴ field is reversing its direction between O and D ∴ the field must be zero somewhere in this region.

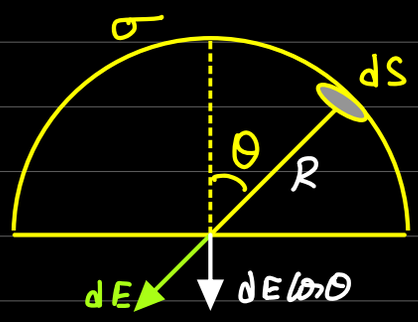
→ Just as apart from centroid we have predicted existence of one more point of zero field on this median, similarly there will be points on other medians as well.

→ In total there are four points in the triangle where the field is zero as shown.



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Field At Centre Of Curvature Of Uniform Hemispherical Shell:



Consider a small patch on surface of a uniformly charged hemispherical shell having a uniform surface charge density σ. Due to the patch dS

$$dE = k\sigma dS/R^2.$$

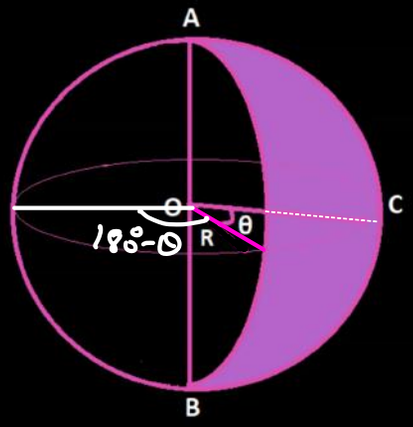
By symmetry we know that radial components of field due to all such patches will cancel out and we will be only left with axial components. Now elemental axial field:

$$dE_a = dE\cos\theta = \frac{k\sigma}{R^2} dS\cos\theta.$$ Now $dS\cos\theta$ is nothing but shadow of patch dS on base of hemisphere due to a beam $\perp$ to the base. Thus $dE_a = \frac{k\sigma}{R^2} dS_{shadow}$ $\Rightarrow$

$$E_a = \frac{k\sigma}{R^2} S_{shadow} = \frac{1}{4\pi\epsilon_0} \times \frac{\sigma}{R^2} \times \pi R^2 = \frac{\sigma}{4\epsilon_0}$$

$\Rightarrow$ $$\boxed{E_H = \frac{\sigma}{4\epsilon_0}}$$

Field At Centre Of Curvature Of Uniformly Charged Lune:

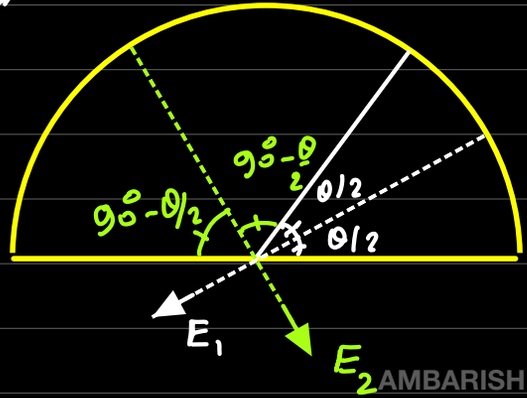


Consider a lune of a sphere having a uniform charge density σ. Now by adding some extra portion we can complete a hemisphere for this purpose, the required lune will have an angle $180°-\theta$. Now we know that field at centre of curvature of hemisphere is $\sigma/4\epsilon_0$. Let the field due to our lune be E_1 and E_2 be the field due to supplementary lune then:

$$E_1 \sin(\theta/2) + E_2 \sin(90°-\theta/2) = \sigma/4\epsilon_0 \quad (1)$$

and

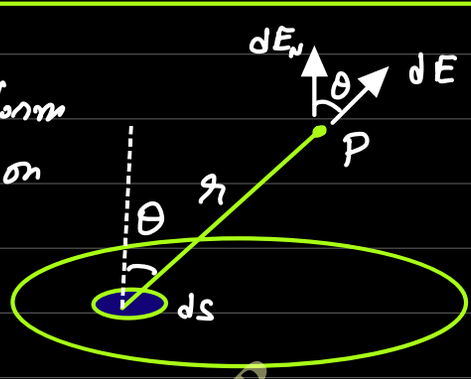
$$E_1 \cos(\theta/2) = E_2 \cos(90°-\theta/2) \quad (2)$$



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Solving ① & ②, $\boxed{E_1 = \frac{\sigma}{4\epsilon_0}\sin(\theta/2)}$

Normal Field Of A Uniformly Charged Lamina Subtending Solid Angle Ω



Consider an arbitrary lamina carrying a uniform charge density σ. Let dS be a small patch on it. Due to this patch, Electric field at point P is given as:

$$dE = \frac{k\sigma\, dS}{r^2} \Rightarrow dE_N = \frac{k\sigma\, dS}{r^2}\cos\theta$$

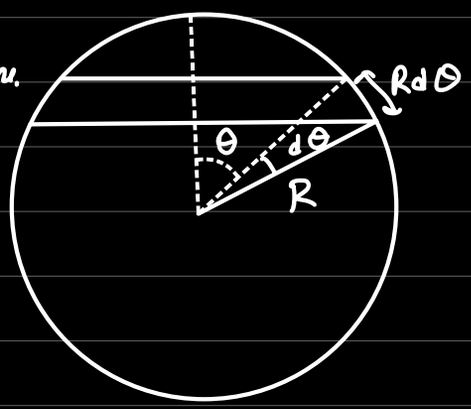
$\Rightarrow dE_N = k\sigma\, d\Omega$ where $d\Omega$ is solid angle subtended by the patch at P.

$\Rightarrow$

$$\boxed{E_N = k\sigma\Omega}$$

Solid Angle Of Cap Of Sphere At Centre Of Curvature :

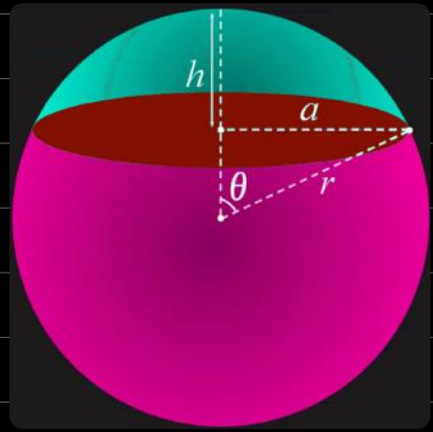
We wish to find the solid angle of cap of a sphere of semivertex angle θ at centre of curvature. Recall that solid angle is shadow area on a spherical screen divided by square of radius of the sphere. Imagine a cap of sphere at angle θ and angular width dθ. If a spherical screen is placed just outside this cap (hugging it) then shadow area $dA = 2\pi R\sin\theta \times R\,d\theta \Rightarrow d\Omega = dA/R^2$



$$\Rightarrow d\Omega = 2\pi\sin\theta\, d\theta \Rightarrow \Omega = \int_0^{\theta} 2\pi\sin\theta\, d\theta$$

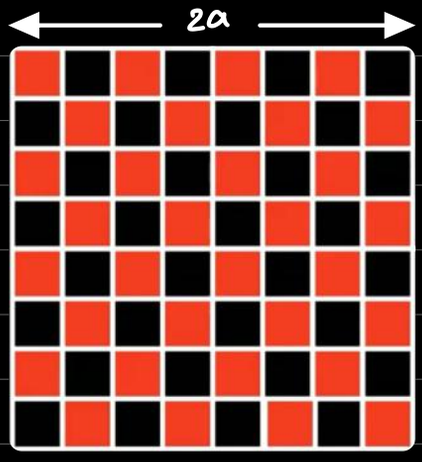
$$\Rightarrow \boxed{\Omega = 2\pi(1-\cos\theta)}$$

- For Hemisphere, $\theta = 90^\circ \Rightarrow \Omega = 2\pi$ Steradians
- For Sphere, $\theta = 180^\circ \Rightarrow \Omega = 4\pi$ steradians.
- Solid angle for the cap is same as the solid angle of the disc "lid" covering it.
- For any two surfaces mounted on same perimeter, solid angle is same.



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→ For a completely closed surface solid angle at interior point is always 4π steradians.



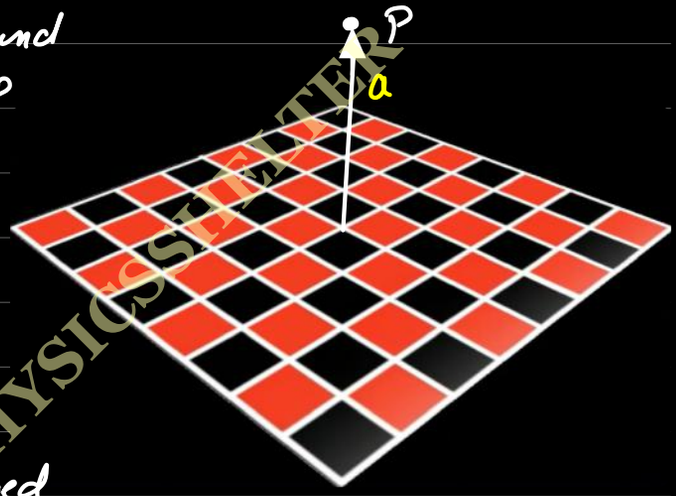
Example: Each red square in the shown chess board has a charge density σ while black squares are uncharged. Find electric field at a point a height 'a' above the centre of the chess board.

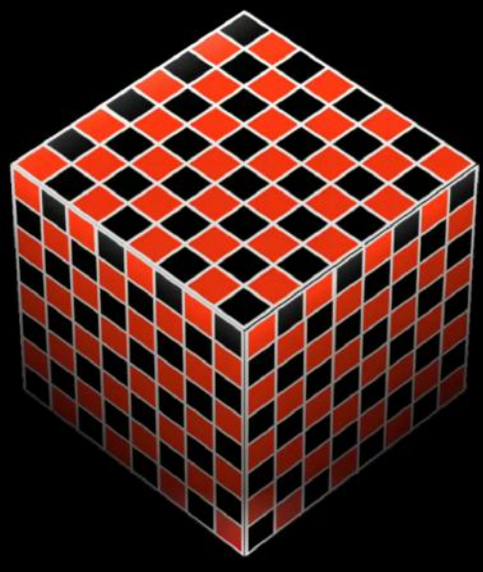
Solution Placing 6 such chess boards around point P symmetrically around point P will complete a cube and hence each chessboard subtends an angle $4\pi/6$ at point P. Thus Ω for each chess board at point P:

$$\Omega_{CB} = 2\pi/3$$

However not entire chess board is charged only 50% of it is charged $\therefore$ solid angle subtended by the charged portion:

$$\Omega_{\sigma} = (2\pi/3)/2 = \pi/3.$$





Now recall the relation between the normal component of field solid angle and charge density as:

$$E_N = k\sigma\Omega$$

$\Rightarrow$ Over our required geometry:

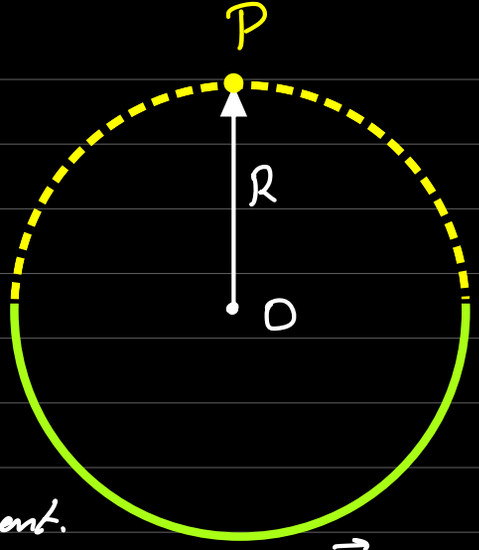
$$E = E_N = k\sigma \times \frac{\pi}{3} = \frac{1}{4\pi\varepsilon_0} \cdot \sigma \times \frac{\pi}{3} = \frac{\sigma}{12\varepsilon_0} \quad \text{Ans.}$$

(Note that by symmetry field component parallel to board vanishes).

Think Yourself: What would be the field if there were three chessboards (See the three visible faces of the last figure above).

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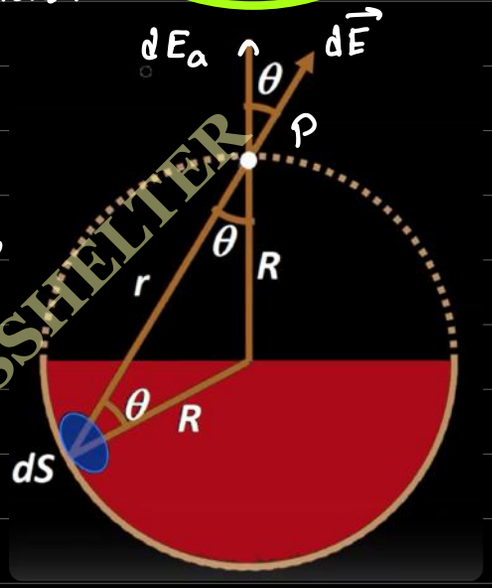
EXAMPLE: A hemispherical shell carries a uniform charge density σ. Find electric field due to this at point P which is vertically above O as shown.



Solution: Consider Electric field produced by a patch on the hemispherical shell of area dS. Small amount of field produced by this element.

$$dE = \frac{k\sigma dS}{R^2}$$, however by symmetry only axial components of elemental fields matter because radial fields will cancel out due to symmetry. Now $dE_a = dE\cos\theta$

$$\Rightarrow \quad dE_a = \frac{k\sigma dS\cos\theta}{R^2} = k\sigma d\omega$$



where $d\omega$ is the solid angle subtended by the patch. Thus total field:

$E_a = k\sigma\Omega$ where Ω is the solid angle subtended by the hemispherical shell at P which is same as solid angle subtended by its disc "lid". The disc lid subtends a semivertex angle 45° at point P. Thus solid angle:

$$\Omega = 2\pi(1-\cos 45^\circ) = 2\pi\left(1-1/\sqrt{2}\right)$$

∴ Required field:

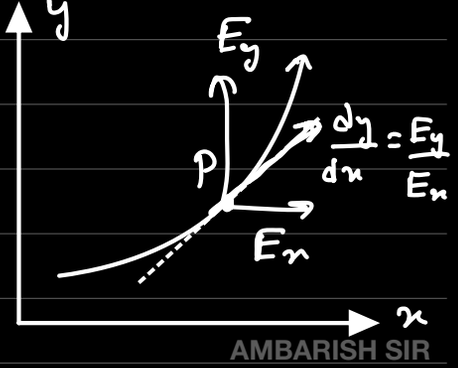
$$E = E_a = \frac{1}{4\pi\epsilon_0}\sigma \times 2\pi\left(1-\frac{1}{\sqrt{2}}\right) = \frac{\sigma}{2\epsilon_0}\left(1-\frac{1}{\sqrt{2}}\right)$$

$$\Rightarrow \quad E = \frac{\sigma}{2\epsilon_0}\left(1-\frac{1}{\sqrt{2}}\right) \text{ Ans.}$$

WRITING EQUATION OF A FIELD LINE FROM THE FIELD FUNCTION:

Let there be a point P on a field line in the x-y plane whose value at a general (x,y) is given as $\vec{E} = E_x(x,y)\hat{i} + E_y(x,y)\hat{j}$ then the slope of this field line is given as:

$$dy/dx = E_y/E_x.$$



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This gives us a differential equation in family of field lines and can be appropriately integrated to get the actual family. The arbitrary constant of integration can be evaluated to find a specific field line passing through a particular point by satisfying the point.

EXAMPLE Field in a certain region in space is given as $\vec{E} = ax\hat{i} + by\hat{j}$. Find the family of field lines. Hence also find the field line passing through point (1,1)

Solution: $dy/dx = E_y/E_x = by/ax$

$$\Rightarrow \int \frac{dy}{by} = \int \frac{dx}{ax} \Rightarrow \frac{1}{b}\ln y = \frac{1}{a}\ln x + C$$

$$\Rightarrow \frac{y^a}{x^b} = \text{Constant}$$, For passing through (1,1),

$$\text{Constant} = \frac{1^a}{1^b} = 1 \Rightarrow \text{required field line is } y^a = x^b$$

or $y = x^{b/a}$. If a & b are positive integers and $b = 2a$ then this becomes a family of parabolae.

For instance adjacent figure shows a field line plot for the field:

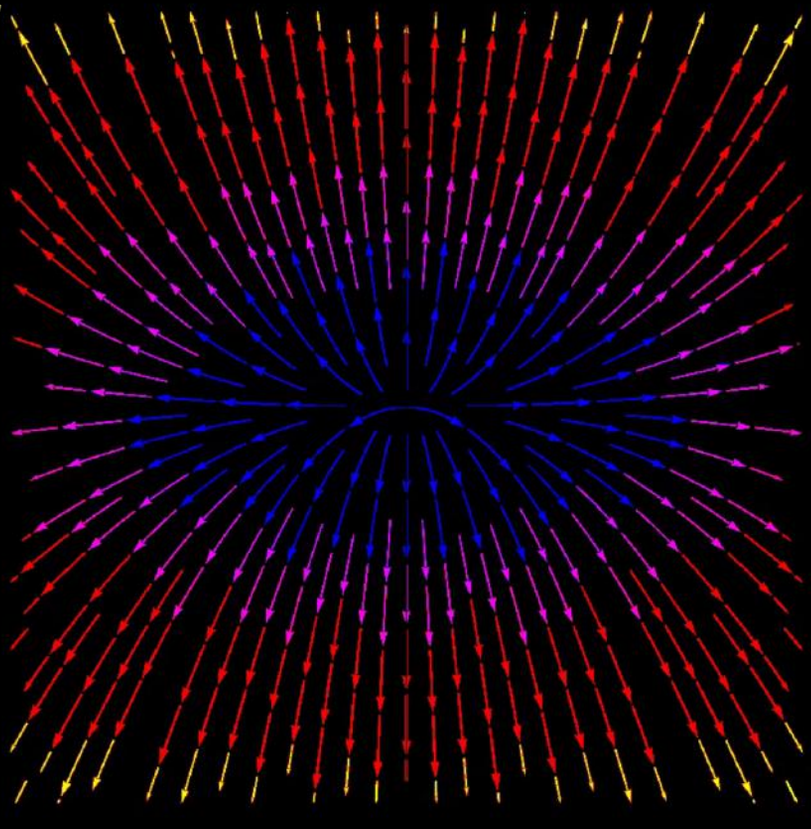
$$\vec{E} = x\hat{i} + 2y\hat{j}$$

Here $\frac{dy}{dx} = \frac{E_y}{E_x} = \frac{2y}{x}$

$$\Rightarrow \frac{dy}{y} = 2\frac{dx}{x}$$

$$\Rightarrow \ln y = 2\ln x + c$$

$\Rightarrow y = Cx^2$. As can be readily seen, this is a family of parabolae as plotted.



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